# Post-Launch Adverse Events Reported to FDA Adverse Event Reporting System for Long-Acting Injectable Buprenorphine

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## BACKGROUND

- Two long-acting injectable buprenorphine (LAIB) medications are approved for the treatment of moderate to severe opioid use disorder (OUD): RBP-6000 (monthly) and CAM2038 (weekly and monthly).[1,2]
- Randomized controlled trials have demonstrated the safety of LAIBs.[3,4] However, post-marketing data provide valuable insights into the safety of treatments within real-world populations.
- FDA Adverse Event Reporting System (FAERS) is a centralized public database used for post-marketing drug safety surveillance.[5]
- Adverse event (AE) rates from FAERS for RBP-6000 and CAM2038 have not been previously analyzed.

## METHODS

- This retrospective analysis evaluated AEs spontaneously reported to FAERS during the 12-month periods following the US commercial launches of RBP-6000 (March 1, 2018 to February 28, 2019) and CAM2038 (September 5, 2023 to September 4, 2024).
  - AEs reported in ≥5% of all case reports for each product were analyzed.
- AE reporting rates (RR) were calculated based on the total number of AEs per 1000 patients treated ($RR_{pt}$) and per 1000 units sold ($RR_{rx}$).
  - The total number of patients treated with/units sold of RBP-6000 and CAM2038 during these periods was estimated based on publicly-reported figures and specialty pharmacy and distributor data.
- AE terms with an RR ≥1 for at least one of the products are reported.

## RESULTS

- The estimated number of patients treated with RBP-6000 and CAM2038 during these periods was 6,464 and 24,805, respectively, with an estimated 16,770 and 94,826 units sold.

**Table 1. AE Reporting Rates in the First 12 Months Post-launch per 1000 Units Sold ($RR_{rx}$) and per 1000 Patients Treated ($RR_{pt}$) (RR ≥1 for either product)**

| AE Term | RBP-6000 $RR_{rx}$ | CAM2038 $RR_{rx}$ | RBP-6000 $RR_{pt}$ | CAM2038 $RR_{pt}$ |
|---|---|---|---|---|
| Withdrawal Syndrome | 15.1 (13.4, 17.1) | 1.1 (0.9, 1.3) | 39.1 (34.7, 44.2) | 4.2 (3.5, 5.1) |
| Injection Site Pain | 15.0 (13.2, 16.9) | 0.3 (0.2, 0.5) | 38.8 (34.3, 43.8) | 1.3 (0.9, 1.8) |
| Drug Dependence | 3.5 (2.7, 4.5) | 0.2 (0.2, 0.4) | 9.0 (7.0, 11.6) | 0.9 (0.6, 1.3) |
| Injection Site Erythema | 3.0 (2.3, 4.0) | 0.2 (0.1, 0.3) | 7.9 (6.0, 10.4) | 0.7 (0.4, 1.1) |
| Injection Site Discharge | 3.0 (2.3, 3.9) | 0.0 (0.0, 0.1) | 7.7 (5.9, 10.2) | 0.0 (0.0, 0.2) |
| Drug Abuse | 2.6 (1.9, 3.5) | 0.0 (0.0, 0.1) | 6.7 (5.0, 9.0) | 0.1 (0.0, 0.3) |
| Nausea | 2.4 (1.8, 3.3) | 0.2 (0.1, 0.3) | 6.3 (4.7, 8.6) | 0.6 (0.3, 1.0) |
| Therapeutic Response Shortened | 0.7 (0.4, 1.3) | 0.4 (0.3, 0.6) | 1.9 (1.1, 3.2) | 1.6 (1.2, 2.2) |
| Drug Ineffective | 0.2 (0.1, 0.6) | 0.4 (0.3, 0.5) | 0.6 (0.2, 1.6) | 1.4 (1.0, 2.0) |

**Figure 1. AE Reporting Rates in the First 12 Months Post-launch per 1000 Units Sold (RR ≥1 for either product)**

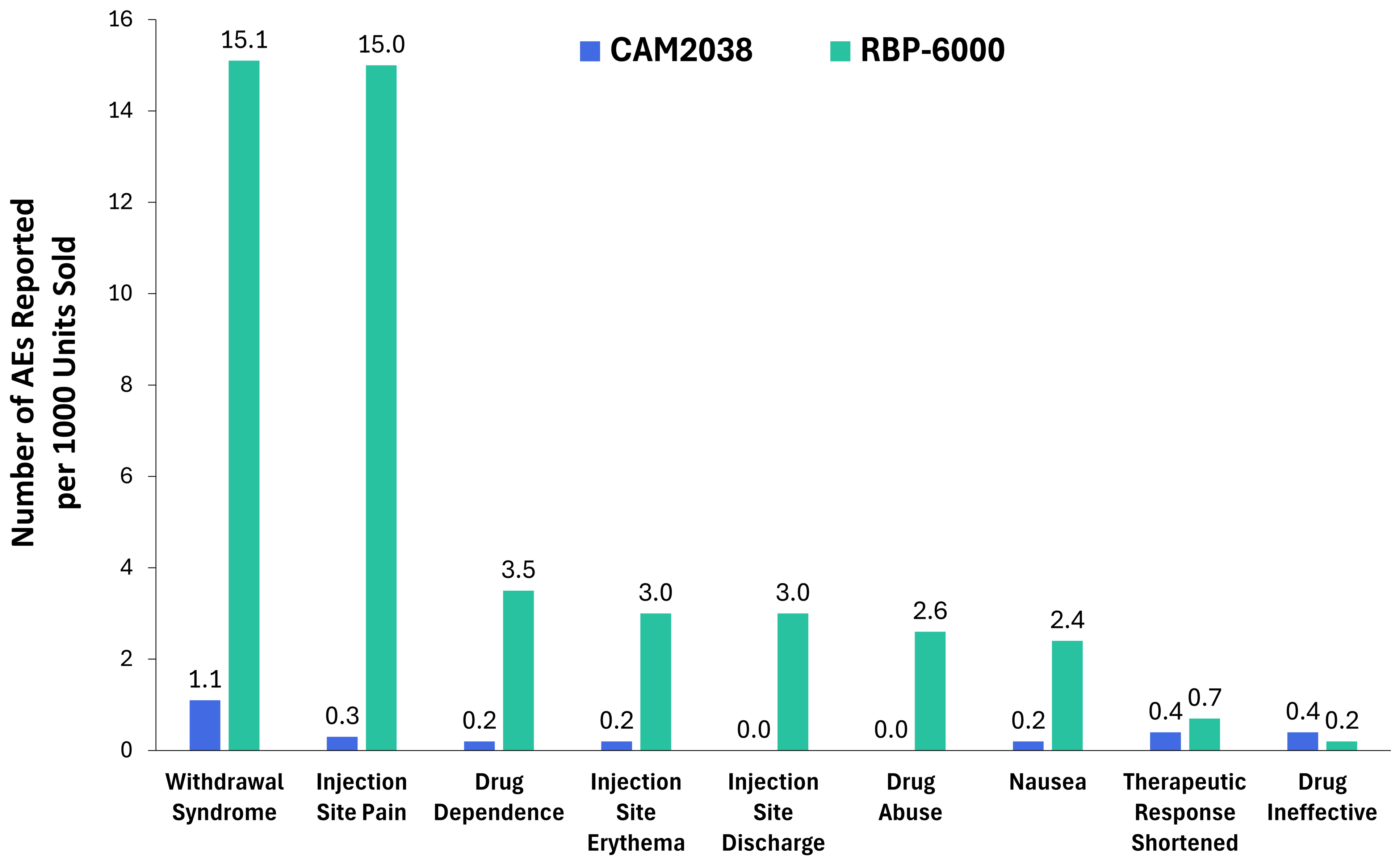
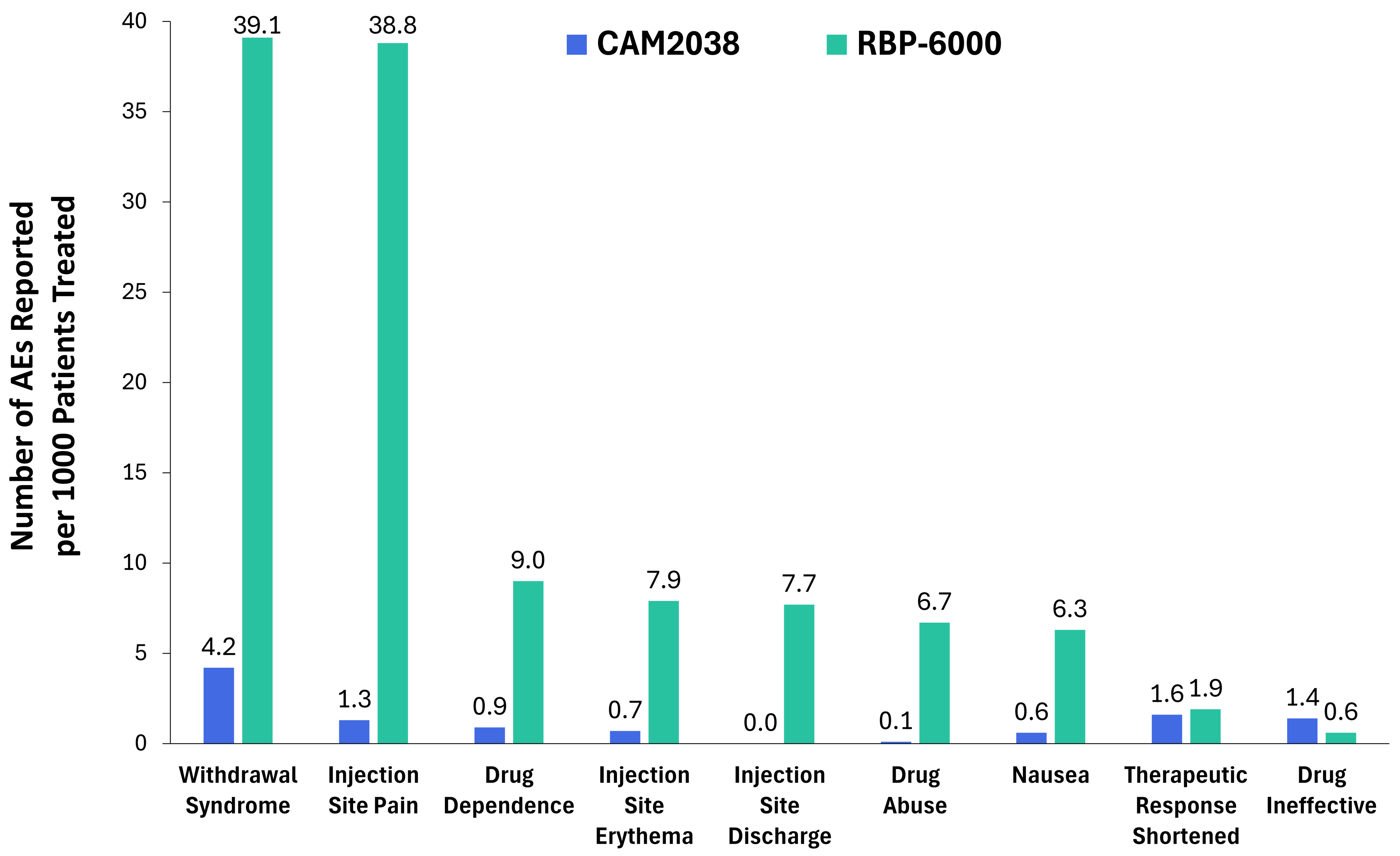



**Figure 2. AE Reporting Rates in the First 12 Months Post-launch per 1000 Patients Treated (RR ≥1 for either product)**




## LIMITATIONS

- There is no certainty that reported AEs were caused by RBP-6000 or CAM2038, given the spontaneous and inconsistent manner in which cases are submitted to FAERS.[5]
- Some AEs related to these medications may not be reported.[5]
- These data may be impacted by the "Weber effect," which is the tendency for increased AE reporting during the initial period following a drug's approval.[6]
  - However, it has been suggested that the Weber effect is not generally observed for AE reporting to FAERS.[6]
- The total number of patients treated with/units sold of RBP-6000 and CAM2038 in the 12-month periods following their US commercial launches was estimated using publicly-reported figures and specialty pharmacy and distributor data, which may be limited by incomplete data capture, duplication, and lack of direct linkage to drug administration.

## CONCLUSIONS

- Overall, AEs reported for both RBP-6000 and CAM2038 were infrequent during the 12-month periods following their US commercial launches.
- These data provide insights into the real-world safety profile of LAIB medications for the treatment of moderate to severe OUD and may help inform clinical decision-making.

## REFERENCES

1. U.S Food & Drug Administration. FDA News Release. FDA approves first once-monthly buprenorphine injection, a medication-assisted treatment option for opioid use disorder. November 30, 2017. https://www.fda.gov/news-events/press-announcements/fda-approves-first-once-monthly-buprenorphine-injection-medication-assisted-treatment-option-opioid.
2. U.S Food & Drug Administration. FDA News Release. FDA Approves New Buprenorphine Treatment Option for Opioid Use Disorder. May 23, 2023. https://www.fda.gov/news-events/press-announcements/fda-approves-new-buprenorphine-treatment-option-opioid-use-disorder.
3. Haight BR, Learned SM, Laffont CM, et al. *Lancet.* 2019;393(10173):778-790.
4. Lofwall MR, Walsh SL, Nunes EV, et al. *JAMA Intern Med.* 2018;178(6):764–773.
5. US Food & Drug Administration. Questions and answers on FDA's Adverse Event Reporting System (FAERS). Accessed April 11, 2025. https://fis.fda.gov/extensions/FPD-FAQ/FPD-FAQ.html
6. Hoffman KB, Dimbil M, Erdman CB, et al. *Drug Saf.* 2014;37(4):283-294.

## DISCLOSURES

BJD received speaking honoraria from Braeburn Inc. MPF received consulting/speaking honoraria from Braeburn Inc, received consultant fees from Molteni Farmaceutici and Accord Healthcare, and is a shareholder of Pocket Naloxone. NRB-K, AF, and JMC are current employees of Braeburn Inc.